

Training Work Instructions: 12” Disruptor Tube Lay Up Part 438

PL2 Doc 13

Objective

The objective of this task is to lay material onto a mandrel to form the composite product

Parts and Tools

Part #	Quantity	Description
440	1	Zyvax Flex-Z 6.0 Release Agent
DM1-3	1	Disruptor Mandrel
TR1-3	1	Oven Trolley
301j	1	Shell Twill, inner mold line
542f	4	Shell 0° Uni layer
542h	4	31”x 36 7/8” piece 90° Uni layer
542i	2	36” x 36 7/8” piece 90° Uni layer
301k	1	Shell Twill, outer mold line
397	1	Surgical Scissors
263	1	Oven Mitts

- **PPE (Personal Protective Equipment)**

- Latex gloves
- Steel toe shoes
- Oven Mitts

Preparation

1. Ensure you have the correct traveler.
2. Apply release agent to the mandrel.
3. Verify you have all the necessary parts in your lay up packet.

Process

1. Place the mandrel centered on the designated lay up table against the backstop.
2. Ensure the ambient temperature is 70-75. If not turn on the heat or AC to change to ideal temperature
3. Ensure the mandrel and table temperature are at 70-80. If not, apply heat to ensure best layup results.
4. Place the first layer of twill flat on the table approximately 1" in front of the mandrel.
 - Ensure the fabric is centered on the mandrel with equal spacing on both ends.
5. Roll the mandrel across the layer of twill to the opposite backstop of the table then return the mandrel by rolling it to the original starting position.
6. Using a black sharpie, mark the seam location of the first layer on the rim of the mandrel.
7. Turn the mandrel so the seam mark is flush on the table.
8. Place the seam reference chart so the P1 line is even with the seam mark on the mandrel.
9. Place the uni shell on the table at the P2 mark on the reference chart.
10. Roll the mandrel across the layer of uni per Step 3.
11. Repeat steps 7 & 8 for each uni shell.
12. Place the three sheets of uni (two at 31" and one at 36") at a 90° position on the table in front of the mandrel ensuring there is no overlap or large gap. Line the first sheet up with the P6 reference mark.
13. Roll the mandrel across the layers of 90° uni to the opposite backstop of the table (see step 3).
14. Repeat steps 10-11 for the next layer of 90° uni.
15. Place the layer of twill on the table with the plastic backing remaining on the bottom. Line this piece with the P8 mark on the reference chart.
16. Roll the mandrel over the final layer and back to the original position.
17. Move the mandrel onto the mandrel support carts. *This will require two people.
 - Roll the carts to the CNC lathe.
18. Using the hoist, place the mandrel onto the lathe.
 - This step requires two production technicians.
19. Follow the instructions for applying compression tape.
 - **Only a certified trained operator may use the CNC lathe.**
20. Once the compression tape is properly applied, place the mandrel onto a trolley.
 - This step requires two production technicians.
21. Roll the trolley to the oven area.
 - Place in the oven at 335°F for 105 minutes.
22. Remove the trolley from the oven and roll it to the cooling area. Place containment signage around the trolley.
 - When the mandrel has cooled sufficiently, begin to cut off the compression tape.
23. Once completely cooled, remove the tube from the mandrel.
24. Using the designated method, write the serial # on the tube.
25. Take the tube and its traveler to the finish station.
26. Return the mandrel and trolley to the designated area for the next lay up preparation.

Safety Considerations and Critical Points

- Always wear appropriate **PPE**.
- Use **extreme caution** when handling sharp objects.
- Follow manufacturer instructions for **oven settings** and **safety precautions**.
- Use **extreme caution** when handling the mandrel after baking—it will be very hot.
- Table, mandrel, and ambient temperature **must** be at stated temperatures. This keeps the fabric from sticking to the table and bubbling up on the mandrel.
- **Check every piece** of material to ensure the correct ply and reference sheet placement.

- When peeling the backing of the uni always peel in the direction of the uni, and start with a corner and peel diagonally to the other side then straighten out to finish removing.
- After each ply check for FOD, bubbles, and creases. If FOD, remove. If bubbles push out or score with a razor blade. If creased, shave down with a razor blade. Ensure **all** creases and bubbles are out before starting the 90's.
- **Never roll the tube backwards once roll has begun in applying material.**
- On ply 6 and 7 (90's), peel backing half way, ensure seams between fabric are touching and not overlapping.
- If the temperature of the mandrel and room is not over 75 degrees, a heat gun should be used to heat the 90's material on plys 6 and 7.
- **Ensure all data is captured on the traveler. Traceability is nonnegotiable.**